

Given:

bearing_std = 0.35 (default_value)

initial robot command $u = [\delta_{rot1}, \delta_{trans}, \delta_{rot2}]^T = [0, 10, 0]^T$

initial mean state $\mu_1 = [x, y, \theta]^T = [180, 50, 0]^T$

alphas $\sqrt{\alpha} = [0.05, 0.001, 0.05, 0.01]^T$ squared root of alphas, used for transition noise in action space (M_t)

Calculate

Covariance Q

Given only bearing and signature (id of the landmark)

$Q = \begin{pmatrix} \delta_\phi^2 & 0 \\ 0 & \delta_s^2 \end{pmatrix} = \begin{pmatrix} 0.35^2 & 0 \\ 0 & 0 \end{pmatrix}$ - noise covariance (corresponds to landmark definition)

Covariance R_t .

Transition function:

$$x_t = g(x_{t-1}, u_t, \epsilon_t = 0) = \begin{bmatrix} x_{t-1} + \delta_{trans} \cos(\theta + \delta_{rot1}) \\ y_{t-1} + \delta_{trans} \sin(\theta + \delta_{rot1}) \\ \theta_{t-1} + \delta_{rot1} + \delta_{rot2} \end{bmatrix}$$

Covariance of noise added to action space:

$\epsilon_t \sim N(0, M_t)$

$$M_t = \begin{bmatrix} \epsilon_{\delta_{rot1}} \\ \epsilon_{\delta_{trans}} \\ \epsilon_{\delta_{rot2}} \end{bmatrix} = \begin{bmatrix} \alpha_1 \delta_{rot1}^2 + \alpha_2 \delta_{trans}^2 & 0 & 0 \\ 0 & \alpha_3 \delta_{trans}^2 + \alpha_4 (\delta_{rot1}^2 + \delta_{rot2}^2) & 0 \\ 0 & 0 & \alpha_1 \delta_{rot2}^2 + \alpha_2 \delta_{trans}^2 \end{bmatrix} = \begin{bmatrix} 0.05^2 * 0^2 + 0.001^2 * 10^2 & 0 & 0 \\ 0 & 0.05^2 * 10^2 + 0.01^2 * (0^2 + 0^2) & 0 \\ 0 & 0 & 0.05^2 * 0^2 + 0.001^2 * 10^2 \end{bmatrix} = \begin{bmatrix} 0.0001 & 0 & 0 \\ 0 & 0.25 & 0 \\ 0 & 0 & 0.0001 \end{bmatrix}$$

Jacobians G_t and V_t with respect to x_{t-1} and u_t respectively

$$G_t = \frac{\partial g(x_{t-1}, u_t, \epsilon_t)}{\partial x_{t-1}} \Big|_{\mu_{t-1}, \epsilon_t=0} = \begin{bmatrix} 1 & 0 & -\delta_{trans} \sin(\theta + \delta_{rot1}) \\ 0 & 1 & \delta_{trans} \cos(\theta + \delta_{rot1}) \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -10 * \sin(0 + 0) \\ 0 & 1 & 10 * \cos(0 + 0) \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix}$$

$$V_t = \frac{\partial g(x_{t-1}, u_t, \epsilon_t)}{\partial u_t} \Big|_{\mu_{t-1}, \epsilon_t=0} = \begin{bmatrix} -\delta_{tras} \sin(\theta + \delta_{rot1}) & \cos(\theta + \delta_{rot1}) & 0 \\ \delta_{trans} \cos(\theta + \delta_{rot1}) & \sin(\theta + \delta_{rot1}) & 0 \\ 1 & 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} -10 * \sin(0 + 0) & \cos(0 + 0) & 0 \\ 10 * \cos(0 + 0) & \sin(0 + 0) & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 10 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Jacobian H_t with respect to x_t corresponds to the observation function

$$H_t = \frac{\partial h(x_t)}{x_t} \Big|_{\bar{\mu}_t=g(\mu_{t-1}, u_t)} = \begin{bmatrix} \frac{-(m_{j,x} - \bar{\mu}_{t,x})}{\sqrt{(m_{j,x} - \bar{\mu}_{t,x})^2 + (m_{j,y} - \bar{\mu}_{t,y})^2}} & \frac{-(m_{j,y} - \bar{\mu}_{t,y})}{\sqrt{(m_{j,x} - \bar{\mu}_{t,x})^2 + (m_{j,y} - \bar{\mu}_{t,y})^2}} & 0 \\ \frac{m_{j,y} - \bar{\mu}_{t,y}}{(m_{j,x} - \bar{\mu}_{t,x})^2 + (m_{j,y} - \bar{\mu}_{t,y})^2} & \frac{-(m_{j,x} - \bar{\mu}_{t,x})}{(m_{j,x} - \bar{\mu}_{t,x})^2 + (m_{j,y} - \bar{\mu}_{t,y})^2} & -1 \end{bmatrix}$$

Finally,

$$R_t = V @ M_t @ V^T = \begin{bmatrix} 0 & 1 & 0 \\ 10 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 0.0001 & 0 & 0 \\ 0 & 0.25 & 0 \\ 0 & 0 & 0.0001 \end{bmatrix} @ \begin{bmatrix} 0 & 10 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 0.25 & 0 & 0 \\ 0 & 0.01 & 0.001 \\ 0 & 0.01 & 0.0002 \end{bmatrix}$$